

Sujal Makwana

suajalm@engineering.upenn.edu | +1 (267)262-3107 | github.com/Maksuzs2401 | linkedin.com/in/sujal-makwana

EDUCATION

University of Pennsylvania | Philadelphia, PA

Master of Science in Engineering in Electrical Engineering

Aug 2026 - May 2028 (expected)

L.D. College of Engineering | Ahmedabad, Gujarat.

B. Tech, Electronics and Communications, GPA: 3.5/4.0

Sep 2021 - May 2025

PROFESSIONAL EXPERIENCE

KMT (Krishna Machine Tools)

Jul 2025 - Jul 2026

R&D Engineer (Hardware development)

- Designed an inductive proximity sensor end-to-end: using a Colpitts LC oscillator, LM393 comparator stage, and 24 V NPN/PNP output stage; sensing range 7 mm with adjustable sensitivity.
- Eliminated touch-induced EMI with a custom grounded Faraday shield between the PCB and metal housing.
- Validated reliability on a custom endurance rig: 12 million switching cycles over 5 days of continuous operation with zero failures. Built a 600-unit pilot run, set up assembly and QC.

Indian Institute of Technology Gandhinagar

May 2024 - Jul 2024

Research Intern

- Designed and built an MCU-controlled over-current protection circuit for a DFB laser driver: 16-bit ADC current sensing, latching MOSFET disconnect at a 115 mA threshold, live OLED current readout, with manual reset.
- Engineered the full transition of a benchtop water vapour sensing system into a UAV-mountable payload, integrating the laser controller, GSM module, Raspberry Pi-based data acquisition, and power management into a compact housing.
- Supported field trials of the system on a hexacopter, measuring atmospheric water vapor at 100 ft altitude.

PUBLICATIONS AND PRESENTATIONS

- S. De, **S. Makwana**, H. Meisheri, and A. L. Chakraborty, "A hexacopter-mounted automated 1f WMS system for vertically-resolved measurement of ambient water vapour," 16th Int. Conf. Photonics-2024 (Invited).
- S. De, D. Paul, **S. Makwana**, H. Meisheri, K. T. V. Grattan, and A. L. Chakraborty, "Vehicle-mounted and hexacopter-mounted ambient water vapour measurement using the R1f/ Δ I1 WMS technique applied to a 1392 nm DFB laser," IEEE Sensors Journal (under review).

PROJECTS

- Systolic-Array GEMM Accelerator:** Designed a 16×16 INT8 systolic-array matrix multiplier (256 PEs mapped one-to-one onto DSP48E2 blocks, 32-bit accumulation) with an AXI4-Stream wrapper (256-bit input, 512-bit output); 103-cycle end-to-end latency per tile (16 load, 71 compute, 16 store).
- Closed timing at 200 MHz (zero failing endpoints) with manual Pblock floorplanning; 5.9K LUTs, 256 DSPs (44%).
- Smith-Waterman AXI-Stream Accelerator:** Adapted the GEMM systolic array into a 16×16 Smith-Waterman DNA-alignment engine: 256 PEs compute scores and traceback pointers along anti-diagonal wavefronts, and a 3-stage pipelined max-score finder returns the optimal score and its coordinates in hardware.
- 86-cycle latency per 16-base alignment (0.30 GCUPS sustained, 1.6 peak); timing met at 100 MHz; 15K LUTs, no DSPs or BRAM.
- FPGA-Based CNN Accelerator for ECG Arrhythmia Classification:** Designed the RTL for a streaming INT8 1D-CNN (3 conv layers, dense layer, argmax; 187-sample input, 5 classes): line-buffer convolution, time-multiplexed DSP MAC arrays, streaming max-pool, and saturating requantization. Hardware/software co-design with an ML partner, who trained the model and exported the quantized weights that initialize the on-chip block RAMs.
- Closed timing at 100 MHz post-implementation on Artix-7 (24.9K LUTs, 229 of 240 DSPs, 24 BRAMs); verified module-level and end-to-end in simulation with known test vectors; validation with trained weights in progress.

SKILLS

- Digital / FPGA:** SystemVerilog, Verilog, RTL design, AXI4-Stream, systolic architectures, (Xilinx Artix-7, UltraScale+, ForgeFPGA)
- Analog & Mixed-Signal:** LC/Colpitts oscillators, comparator and hysteresis design, discrete BJT switching stages, EMI

shielding, PCB design

- **Embedded & Programming:** C, Python, Arduino, ESP32-S3, Raspberry Pi
- **Tools:** Vivado, Cadence, KiCad, LTspice, MATLAB

Leadership & Service

- President, IETE student chapter, L.D. College of Engineering: As a president of this forum I led a team of 25 members and organized campus-wide engineering workshops and events for students.
- Volunteer, Vatsalyadham Vruddhashram (old-age home): Active service member who helped organize the home's events,, and supported its digital marketing campaign across social media platforms.